Name: Shubham Agarwal
Class: 25MAM-1 UID: 25MCI10091
Section: 25MAM1-A

PAGE No
DATE

Front-End Assignment

Implementation of State Management using Redux

Introduction to Redux-

Redux is an opensource JavaScript Library for managing and centralizing application state. It is most commonly used with libraries like react or angular for building UI.

→ using react and passing data through components (prop drilling) can become complex. Redux solve this by providing a "Single Source of Truth."

Why use Redux —

→ Easier to manage data flow throughout the application

Core Components of Redux

- Store: Think of it as the "Brain". It holds the entire state of the application. There is one store in redux.

- Actions: These are plain JavaScript objects that represents "what user wants to do. Every action must have property.

- Reducers: These are pure functions that decide how the state will change. It takes current states and an action as an arguments and returns new state.

- Dispatch: This is the function used to send an action to the reducer. It is the only way to trigger state change.

PAGE No
DATE

Implementation : Simple counter manager app -

Step 1: Define Actions

```
// exporting from actionType.js
export const INCREAMENT = "INCREAMENT"
export const DECREMENT = "DECREMENT"

// imparting in action.js

impart { INCREMENT, DECREMENT } from './actionType';
export const incrementCount = () => ({type : INCREMENT});

export const decrementCount = () => ({type : DECREMENT});
```

Step 2: Create the Reducer

```
import { INCREMENT, DECREMENT } from './actionType';
const initialState = { count : 0 };

const counterReducer = { state = initialState, action ) =>
{ switch (action.type) {
      case INCREMENT :
        return { ...state , count : state.count +1 };
    case DECREMENT :
      return {...state, count. count -1 };
    default :
      return state;
  }
};
export default counterReducer;
```

Step 3: Component Integration

```
// Counter.js

import React from 'React';
import { useSelector, useDispatch } from 'react-redux';
import { incrementCount, decrementCount } from ./action';

const Counter = () => {
  const count = useSelector ((state) => state.count);
  const dispatch = useDispatch();

  return (
    <div style={{ textAlign: 'Center', marginTop: '50px' }}>
      <h1> Counter : {count}</h1>
      <button onClick = {() => dispatch (incrementCount())}>
        Increment </button>
      <button onClick = {() => dispatch (decrementCount())}>
        Decrement </button>
    </div>
  );
};
```